

IDENTIFICATION

Product Code: MAINDEC-08-D3EB-D
Product Name: TC01 Extended Memory Exerciser
Date Created: January 5, 1968
Maintainer: Diagnostic Group
Author: Edward P. Steinberger

1. ABSTRACT

TC01 EXTENDED MEMORY EXERCISER is a test program for the PDP-8 Computer which tests the transfer to data between the TC01 DECTape Control and extended memory fields (more than 4K). It does this by storing a data pattern in an extended memory field, transferring the data onto DECTape and then reading the data back into the field and checking it for correct transfer.

2. REQUIREMENTS

2.1 Equipment

Standard PDP-8 Computer

TC01 DECTape Control with at least 1 Transport (TU55)

183 Memory Extension Control with at least 1 Memory Module (184)

2.2 Storage

The program occupies the first 6 pages of Bank 0 and uses 2000 to 5777 of each memory bank for data storage. All of memory not occupied by the program in Bank 0 with the exception of the last page is filled with "HLT".

2.3 Preliminary Programs

DECTape Basic Exerciser

DECTape Random Exerciser

3. LOADING PROCEDURE

3.1 Method

This test is loaded by the standard Binary Loader (SA = 7777).

4. STARTING PROCEDURE

4.1 Control Switch Settings

The following is a table of control switch settings and their action on the program.

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SR	Set As	Action
0 }		Unit Select bits for DECtape transport
1 }		
2 }		
6 }		Number of ADDITIONAL memory fields (must be non-zero)
7 }		
8 }		
9	1	Halt on Error
	0	Don't Halt on Error
10	1	Don't Print Errors
	0	Print Errors
11	1	Don't Ring Bell on Error
	0	Ring Bell on Error

4.2 Starting Address

The starting address of the program is 00200.

4.3 Program and/or Operator Action

4.3.1 Load program into Memory Bank 0 per 3.1.

4.3.2 Set SR to 00200, depress "Load Address".

4.3.3 Set SR 9 to 11 per 4.1.

4.3.4 Depress "Start".

5. OPERATING PROCEDURE

5.1 Operational Switch Settings

See 4.1

5.2 Subroutine Abstracts

None

5.3 Operating Procedure

After starting the program per 4.3 the computer will halt at location 00223 if no error occurred after performing static tests on the TC01 "Field" register.

5.3.1 Set SR 0 to 2 to unit select bits of transport to be exercised.

5.3.2 Place a standard PDP-8 certified DECtape on the transport to be exercised, place transport "On Line" with "Write" enabled.

5.3.3 Set SR 6 to 8 to the number of extra memory fields (non-zero).

5.3.4 Depress "Continue".

5.3.5 To run the dynamic tests only:

5.3.5.1 Set SR to 0224, depress "Load Address".

5.3.5.2 Set unit select bit, extra field bits, error option bits in SR (see 4.1).

5.3.5.3 Assure selected transport is ready.

5.3.5.4. Depress "Start".

6. ERRORS

6.1 Error Halts and Description

The following is a table of error halts and the reason for each.

Location	Reason
0227	No extended memory indicated by SR 6 to 8
0351 (HALT 2)	"B" register not properly set
0527 (HALT 3)	Data Error
0735 (HALT 1)	DECtape Error
Outside of Program	Extended Memory Control Error (either non-existent or defective memory)

6.2 Error Recovery

6.2.1 Reset SR if necessary.

6.2.2 Depress "Continue" for any error except "Outside of program".

6.3 Error Typeouts

6.3.1 "B" Register Error.

MEMORY FIELD ERROR

RIGHT	WRONG
0070	0030

The above example shows that an attempt was made to set the "B" register to 0070, however the most significant bit (0040) did not set.

6.3.2 Data Error

DATA ERROR
FIELD 0003
FIRST BLOCK 0040
LOC. DATA
2000 7402
2001 7402

The above example shows that a data error occurred in Memory Bank 3, the transfer started at block 0040, location 2000 contains 7402 (should contain 2000).

6.3.3 DECtape Error

THE FOLLOWING UNEXPECTED ERRORS OCCURRED:
MARK TRACK
END ZONE
SELECT
PARITY
TIMING

The above typeout (with at least one error indicated) will be typed out if there is a DECtape control error.

7. RESTRICTIONS

7.1 Starting Restrictions

None

7.2 Operating Restrictions

SR6 to 8 may be set to less than the number of additional memory fields but not more than that number. (SR6 to 8 must be non-zero), otherwise unpredictable results may occur (attempts to reference non-existent memory).

8. MISCELLANEOUS

8.1 Execution Time

Not applicable - 1 Pass down tape allows each memory field (other than 0) to be exercised at least 34 (Dec.) times (takes 8 minutes).

9. PROGRAM DESCRIPTION

9.1 The first portion of the test performs static tests on the memory field portion of the "B Register". The "B Register" is tested to assure that it may be set to all values (0 to 7). Any error will cause an error timeout and error halt unless these are suppressed by Switch Register settings.

9.2 The second portion of the test performs dynamic tests on the DECtape control, transfers are made to and from DECtape and extended memory.

9.2.1 The program first obtains the maximum field size from SR 6 to 8 and checks to make sure it is non-zero. The program then extracts the unit select bits from SR 0 to 2 for the DECtape drive being exercised.

9.2.2 The program then sets a location so that the first block sought is block 0 ("current block").

9.2.3 The program then sets a location so that field 1 is exercised ("current memory field").

9.2.4 The "current memory field" is then checked to assure that it is not larger than the maximum available field. If it is larger, the program goes to 9.2.3, otherwise the program goes to 9.2.5.

9.2.5 "HLT" is stored in all memory locations in field 0 not occupied by the program or the Binary or Rim Loaders. Also a location in an error typeout routine is initialized to provide error header typeout.

9.2.6 "HLT" is stored in all memory locations in the "current memory field", then data (addresses) are stored in locations 2000 through 5777 of the "current memory field".

9.2.7 The "current block" is then searched for. If a DECTape error occurs, an error typeout occurs and the search process is repeated.

9.2.8 After the "current block" has been found, the data in the "current memory field" is written on DECTape starting at that block. If an error occurs, the program goes back to 9.2.7. otherwise it goes to 9.2.9.

9.2.9 All locations in the "current memory field" are then set to "HLT".

9.2.10 The "current block" is sought again.

9.2.11 The data just written on DECTape is then read back into "current memory field" at the locations from which it came. A DECTape error at this point returns the program to 9.2.10.

9.2.12 The data in the "current memory field" is then checked to assure correctness of transfer.

9.2.13 All locations in the "current memory field" are set to "HLT".

9.2.14 The "current block" is incremented by 10 and checked to assure that it does not equal 2670. If it does, the "current block" is then set back to 0.

9.2.15 The "current memory field" is then incremented by 10 (effectively 1) and the program goes back to 9.2.4.

/PROGRAM TO EXERCISE THE IC01 AND EXTENDED MEMORY

0020

*20

/CONSTANTS AND VARIABLES

0020	0000	BLOCK,	0	/CURRENT BLOCK
0021	0000	CNTR,	0	
0022	0000	ERROR,	0	/ERROR STATUS
0023	0000	FIELD,	0	/CURRENT FIELD
0024	0002	K0002,	2	
0025	0003	K0003,	3	
0026	0004	K0004,	4	
0027	0007	K0007,	7	
0030	0010	K0010,	10	
0031	0070	K0070,	70	
0032	0130	K0130,	130	
0033	0150	K0150,	150	
0034	0200	K0200,	200	
0035	0201	K0201,	201	/MINUS 75//
0036	0207	K0207,	207	/BELL
0037	0212	K0212,	212	/LF
0040	0215	K0215,	215	/CR
0041	0240	K0240,	240	/SPACE
0042	0260	K0260,	260	/DIGIT CODE
0043	0400	K0400,	400	/FWO-REV
0044	0600	K0600,	600	/GO-REV
0045	0610	K0610,	610	/GO REV SEARCH
0046	1777	K1777,	1777	/FIRST ADDRESS-1 OF DATA
0047	2670	K2670,	2670	
0050	4000	K4000,	4000	/NUMBER OF DATA WORDS
0051	7000	K7000,	7000	
0052	7401	K7401,	7401	/MINUS RUBOUT
0053	7754	K7754,	7754	/KC
0054	7755	K7755,	7755	/CA
0055	7774	K7774,	7774	/MINUS 4
0056	0000	MAX,	0	/HIGHEST FIELD AVAILABLE
0057	1000	PMESS1,	MESS1	/DECTAPE ERROR HEADER
0060	1056	PMESS2,	MESS2	/MARK TRACK
0061	1075	PMESS3,	MESS3	/END ZONE
0062	1106	PMESS4,	MESS4	/SELECT
0063	1117	PMESS5,	MESS5	/PARITY
0064	1150	PMESS6,	MESS6	/TIMING
0065	1141	PMESS7,	MESS7	/"B" REGISTER ERROR HEADER
0066	1205	PMESS8,	MESS8	/DATA ERROR HEADER
0067	1232	PMESS9,	MESS9	/MORE DATA ERROR HEADER
0070	1251	PMES10,	MESS10	/END OF DATA ERROR HEADER
0071	0000	PNTK1,	0	/MSPHNT POINTER
0072	0000	PNTK2,	0	/DATERR POINTER
0073	7410	SKIP,	SKP	
0074	7402	STOP,	HLT	
0075	0000	TEMP,	0	
0076	0000	UNIT,	0	/UNIT BEING OPERATED UPON
0077	0400	Z1,	SET	
0100	0416	Z2,	STORE	

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0101 0437
0102 0466
0103 0000
0104 0051
0105 1266

23, CHECK
24, DATERK+4
25, SEARCH
26, WAIT
27, END

/MESSAGE PRINT SUBROUTINE

```

106 0000      MSPRNI, 0
107 3071      DCA PNTR1
110 1471      TAD I PNTR1
111 4117      JMS TYPE
112 1052      TAD K/401
113 7050      SNA CLA
114 5506      JMP I MSPRNI
115 2071      JSZ PNTR1
116 5110      JMP MSPRNI+2

117 0000      TYPE, 0
120 0046      TLS
121 0041      TSF
122 5121      JMP ,71
123 5517      JMP I TYPE

124 0000      CRLF, 0
125 7200      CLA
126 1040      TAD K0215
127 4117      JMS TYPE
130 7200      CLA
131 1037      TAD K0212
132 4117      JMS TYPE
133 7200      CLA
134 5524      JMP I CRLF

```

/PRINT SUBROUTINE

```

135 0000      PRINT, 0
136 7004      LAS
137 7012      RIR
140 7030      S&L CLA
141 5535      JMP I PRINT
142 2135      JSZ PRINT
143 5535      JMP I PRINT

```

/BELL SUBROUTINE

```

144 0000      BELL, 0
145 7004      LAS
146 7010      RAR
147 7030      S&L CLA
150 5544      JMP I BELL
151 1036      TAD K0207
152 4117      JMS TYPE
153 5544      JMP I BELL

```

/TYPE OUT THE NUMBER IN THE AC

0134	0000	NUMBER, 0
0135	3075	DCA TEMP
0136	1055	TAO K774
0137	3021	DCA CNTR
0160	1075	TAO TEMP
0161	7104	RAL CLL
0162	7004	RAL
0163	7006	RIL
0164	3075	DCA TEMP
0165	1075	TAO TEMP
0166	0027	AND K0007
0167	1042	TAO K0200
0170	4117	JMS TYPE
0171	7200	CLA
0172	1075	TAO TEMP
0173	2021	ISE CNTR
0174	5102	JMP +12
0175	7200	CLA
0176	5554	JMP I NUMBER

6761	DTXA=0761
6762	DTCA=0762
6764	DTXA=0764
6766	DTLA=0766
6771	DTST=0771
6772	DTWB=0772
6774	DTLB=0774
6201	CDF=6201

0200

*200

/STATIC - SET "B" TESTS AND READ BACK

```

1200 1300      BEGIN, CLA CLL
1201 3023      DCA FIELD      /CLEAR FIELD
1202 6774      DILB          /LOAD "B"
1203 1200      CLA
1204 6772      DIRB          /READ "B"
1205 0031      AND K0070
1206 3056      DCA MAX        /AND SAVE
1207 1056      TAD MAX
1210 1041      CIA
1211 1023      TAD FIELD
1212 1040      SEA CLA        /SAME AS NUMBER SET?
1213 4327      JMS BERRQR     /NO, ERROR
1214 1023      TAD FIELD
1215 1030      TAD K0010      /INCREMENT FIELD SETTING
1216 0031      AND K0070
1217 3023      DCA FIELD
1220 1023      TAD FIELD
1221 1440      SEA          /DONE ALL FIELDS?
1222 5202      JMP BEGIN+2    /NO
1223 1402      HLT

```

/DYNAMIC TESTS

```

1224 1004      START, LAS
1225 0031      AND K0070      /GET MAXIMUM FIELD SIZE
1226 1450      SNA          /NON-ZERO?
1227 1402      HLT          /NO
1230 3056      DCA MAX      /YES, STORE
1231 1004      LAS
1232 0031      AND K/0000    /GET UNIT NUMBER
1233 3076      DCA UNIT     /AND SAVE
1234 3020      DCA BLOCK    /CLEAR BLOCK
1235 1030      TAD K0010    /SET TO OPERATE
1236 3023      DCA FIELD    /ON FIELD 1
1237 1023      TAD FIELD    /COMPARE CURRENT
1240 1041      CIA          /FIELD AGAINST
1241 1056      TAD MAX      /MAXIMUM FIELD
1242 1110      SPA CLA      /IS CURRENT FIELD TOO LARGE?
1243 5235      JMP ,=6      /YES, RESET TO FIELD 1
1244 4353      JMS HALIS    /STORE HALI IN MEMORY FIELD 0
1245 1073      TAD SKIP
1246 3502      DCA I Z4
1247 1023      TAD FIELD    /STORE HALI IN
1250 4477      JMS I Z1     /MEMORY FIELD "N"
1251 1023      TAD FIELD    /SET INTO FIELD "N"
1252 4500      JMS I Z2     /DATA (ADDRESSES) TO BE WRITTEN ON TAPE
1253 4503      JMS I Z5     /SET UP DECTAPE TO
1254 4566      JMS ERR      /ATTACK BLOCK IN FORWARD DIRECTION
1255 5253      JMP ,=2      /CHECK FOR ERROR
                          /REPEAT SEARCH

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0256 1023      TAD FIELD          /RETURN HERE WHEN BLOCK IS FOUND AND NO ERRORS
0257 6774      DILB              /LOAD MEMORY FIELD REGISTER
0260 1033      TAD K0150
0261 6764      DIXA              /CHANGE FROM SEARCH TO WRITE DATA CONT.
0262 1046      TAD K1777
0263 3454      DCA I K7755      /SET UP CA
0264 1050      TAD K9000
0265 3453      DCA I K7754      /AND WC
0266 4504      JMS I Z0          /WAIT FOR DECIAPPE FLAG AND NO ERRORS
0267 4366      JMS EHR
0270 5253      JMP I-12
0271 1023      TAD FIELD          /SET THE CURRENT MEMORY FIELD TO HLI
0272 4477      JMS I Z1
0273 4503      JMS I Z0          /FIND BLOCK AGAIN
0274 4366      JMS EHR
0275 5273      JMP I-2
0276 1023      TAD FIELD          /SET MEMORY FIELD REGISTER
0277 6774      DILB
0300 1032      TAD K0150          /SEARCH TO READ DATA CONT
0301 6764      DIXA
0302 1046      TAD K1777
0303 3454      DCA I K7755      /SET UP CA
0304 1050      TAD K9000
0305 3453      DCA I K7754      /AND WC
0306 4504      JMS I Z0          /WAIT FOR DECIAPPE FLAG AND NO ERRORS
0307 4366      JMS EHR
0310 5273      JMP I-12
0311 1023      TAD FIELD          /CHECK FOR CURRENT DATA
0312 4501      JMS I Z0
0313 1023      TAD FIELD          /SET IT TO HALT AGAIN
0314 4477      JMS I Z1
0315 1020      TAD BLOCK          /INCREMENT BLOCK
0316 1030      TAD K0010          /BY 10
0317 5020      DCA BLOCK
0320 1020      TAD BLOCK
0321 7041      CIA
0322 1047      TAD K2670
0323 7750      SPA SNA CLA      /END OF TAPE?
0324 5020      DCA BLOCK          /YES, ZERO BLOCK
0325 1023      TAD FIELD
0326 5235      JMP START+11      /RETURN TO TEST NEXT MEMORY FIELD

```

/"B" REGISTER ERROR SUBROUTINE
 BERRON, 0

```

0327 0000      JMS BELL
0330 4144      JMS PRINT
0331 4135      JMP HALT2-5
0332 5346      TAD PMESS7
0333 1065      JMS MSPHNI
0334 4106      TAD FIELD
0335 1023      JMS NUMBER
0336 4154      TAD K0240
0337 1041      JMS TYPE
0340 4117      JMS TYPE
0341 4117

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0342	7200	CLA
0343	1056	TAD MAX
0344	4154	JMS NUMBER
0345	4124	JMS CNLP
0346	7604	LAS
0347	0026	AND K0004
0350	7640	SZA CLA
0351	7402	HAL 12: HLT
0352	5727	JMP I BERRUK

```

0400      0000
0401      7450
0402      5000
0403      1214
0404      5206
0405      5010
0406      6201
0407      1074
0410      5410
0411      1010
0412      7040
0413      5207
0414      6201
0415      5000

*400
/SUBROUTINE TO STORE HALTS IN MEMORY BANK "N" (N-NONZERO), IN AC(6-8)

SET,      0
          SNA
          JMP I SET
          TAD ,+11
          DCA ,+2
          DCA 10
          CUF
          TAD STOP
          DCA I 10
          TAD 10
          SEA CLA
          JMP ,+4
          CUF
          JMP I SET
/SUBROUTINE TO STORE ADDRESSES IN MEMORY BANK "N" (N-NONZERO, IN AC6-8)

STORE,    0
          SNA
          JMP I STORE
          TAD ,+14
          DCA ,+5
          TAD K1777
          DCA 10
          TAD K4000
          DCA 12
          CUF
          TAD 10
          IAC
          DCA I 10
          ISZ 12
          JMP ,+4
          CUF
          JMP I STORE

/SUBROUTINE TO CHECK MEMORY BANK "N" TO ASSURE PROPER DATA STORED

CHECK,    0
          SNA
          JMP I CHECK
          TAD ,+10
          DCA ,+7
          TAD K1777
          DCA 10
          TAD K4000
          DCA 12
          TAD 10
          CMA
          CUF
          TAD I 10
          SEA CLA

```

/SUBROUTINE TO STORE HALTS IN MEMORY BANK 0

353	0000	HALTS:	0
354	0201		CUF
355	1103		TAD Z/
356	3011		DCA 11
357	1074		TAD STOP
360	3411		DCA I 11
361	1011		TAD 11
362	1033		TAD K0201
363	7640		SZA CLA
364	3337		JMP ,=3
365	3733		JMP I HALTS

/DECTAPE ERROR REPEAT TEST SUBROUTINE

366	0000	ERR:	0
367	7200		CLA
370	6772		DIRB
371	7700		SMA CLA
372	2366		ISZ ERR
373	3766		JMP I ERR

/DATA ERROR SUBROUTINE

```

0462 0000      DATERR, 0
0463 4144      JMS BELL
0464 4155      JMS PRINT
0465 5524      JMP HALIS-3
0466 7410      SKP                      /PRINT MESSAGE HEADER?
0467 5512      JMP ,+25                /NO
0470 6201      CUF
0471 1066      TAD PMESS0             /YES, TYPE FIRST PART
0472 4106      JMS MSPRNI
0473 1025      TAD FIELD
0474 7112      CLL RTR
0475 7010      RAR
0476 4154      JMS NUMBER             /TYPE OUT FIELD
0477 1067      TAD PMESSY
0500 4106      JMS MSPRNI             /MORE HEADER
0501 1020      TAD BLOCK
0502 4154      JMS NUMBER             /FIRST BLOCK NUMBER
0503 1070      TAD PMESS10
0504 4106      JMS MSPRNI             /REST OF HEADER
0505 1051      TAD K7000
0506 3266      DCA DATERR+4
0507 1252      TAD CHECK+15          /FORM "CUF"
0510 3511      DCA ,+1
0511 6201      CUF                   /CHANGE FIELD
0512 1010      TAD 10
0513 3072      DCA PNTR2
0514 1072      TAD PNTR2
0515 4154      JMS NUMBER             /TYPE OUT LOCATION
0516 1041      TAD K0240
0517 4117      JMS TYPE               /1 SPACE
0520 7200      CLA
0521 1472      TAD I PNTR2
0522 4154      JMS NUMBER             /TYPE OUT DATA
0523 4124      JMS CNLF              /CNLF
0524 7604      LAS
0525 0026      AND K0004
0526 7640      SZA CLA               /HALT?
0527 7402      HLT                   /YES
0530 5662      JMP I DATERR

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J455 4262
J456 2012
J457 5250
J460 6201
J461 5637

JMS DATERR /DATA ERROR
ISZ 12
JMP , - /
CUF
JMP I CHECK

00000

*0000

/SEARCH SUBROUTINE

```

0000 0000 SEARCH, 0
0001 1200 CLA
0002 3454 DCA I K//55 /BLOCK# TO LOC 0
0003 1076 TAD UNIT /COMBINE UNIT
0004 1045 TAD K0610 /AND SEARCH, NORM, REV
0005 6766 DTLA /LOAD A
0006 6774 DILB /CLEAR B
0007 6771 DIS /WAIT FOR
0010 5207 JMP ,=1 /SOME FLAG
0011 6772 DIRB /READ B
0012 7006 RIL
0013 7700 SMA CLA /END ZONE?
0014 5220 JMP ,=4 /NO
0015 1044 TAD K0600 /YES, TURN.
0016 6764 DIXA /AROUND
0017 5207 JMP SEARCH+ /
0020 6772 DIRB /READ STATUS B
0021 7700 SMA CLA /DECTAPE ERROR?
0022 5225 JMP ,=3 /NO
0023 9251 JMS WAIT /YES, STOP TRANSPORT, ETC.
0024 5201 JMP SEARCH+1 /TRY SEARCHING AGAIN
0025 6761 DIRA /READ A
0026 7006 RIL /MOVE DIRECTION
0027 7006 RIL /BIT INTO LINK
0030 7200 CLA /CLEAR AC
0031 1000 TAD 0 /GET CURRENT BLOCK NUMBER
0032 7041 CIA
0033 1020 TAD BLOCK
0034 7450 SNA /CORRECT BLOCK?
0035 5245 JMP FOUND /YES, CHECK DIRECTION
0036 7041 CIA /NO, TAKE 2'S COMPLEMENT
0037 7420 SNL /LINK IS 1 IF BKWD AND NOT AT OR LOWER THAN BLOCK
0040 1024 TAD K0002 /ADD TWO TO ENABLE TURN AROUND
0041 7020 SNL CLA /TURN AROUND (3 BEYOND)?
0042 1045 TAD K0400 /YES
0043 6764 DIXA /CLEAR FLAG
0044 5207 JMP SEARCH+ /WAIT FOR NEXT FLAG
0045 7020 SNL CLA /FOUND BLOCK FORWARD?
0046 5245 JMP ,=3 /NO
0047 6764 DIXA /YES, CLEAR FLAG
0050 5000 JMP I SEARCH /EXIT
FOUND,

```

/SUBROUTINE TO WAIT FOR DECTAPE FLAG AND NO ERROR
/EXIT WITH TRANSPORT STOPPED

0651	0000	WAIT,	0	
0652	6771		DISP	/WAIT FOR SOME FLAG
0653	5252		JMP ,+1	
0654	6761		DIRA	/READ STATUS A
0655	0034		AND K0200	
0656	1025		TAD K0003	
0657	6764		DIRA	/CLEAR GO
0660	6772		DIRB	
0661	7700		SMA CLA	
0662	5051		JMP I WAIT	
0663	4144		JMS BELL	
0664	4135		JMS PRINT	
0665	5332		JMP HALT-3	
0666	1057		TAD PMESS1	/TYPE OUT ERROR MESSAGE HEADER
0667	4106		JMS MSPRNI	
0670	6772		DIRB	
0671	7006		RIL	
0672	5022		DCA ERROR	
0673	7420		SNL	/MARK TRACK ERROR?
0674	5277		JMP ,+3	/NO
0675	1060		TAD PMESS2	
0676	4106		JMS MSPRNI	
0677	1022		TAD ERROR	
0700	7104		RAL CLL	
0701	5022		DCA ERROR	
0702	7420		SNL	/END ZONE?
0703	5306		JMP ,+3	/NO
0704	1061		TAD PMESS3	
0705	4106		JMS MSPRNI	
0706	1022		TAD ERROR	
0707	7104		RAL CLL	
0710	5022		DCA ERROR	
0711	7420		SNL	/SELECT ERROR?
0712	5315		JMP ,+3	/NO
0713	1062		TAD PMESS4	
0714	4106		JMS MSPRNI	
0715	1022		TAD ERROR	
0716	7104		RAL CLL	
0717	5022		DCA ERROR	
0720	7420		SNL	/PARITY ERROR?
0721	5324		JMP ,+3	
0722	1063		TAD PMESS5	
0723	4106		JMS MSPRNI	
0724	1022		TAD ERROR	
0725	7104		RAL CLL	
0726	7620		SNL CLA	/TIMING ERROR?
0727	5332		JMP ,+3	
0730	1064		TAD PMESS6	

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0731 4106      JMS MSPNNI
0732 7004      LAS
0733 0026      AND K0004
0734 7040      SZA CLA      /HALT ON ERROR?
0735 7402      HALT1, HLT
0736 5051      JMP I WAIT
```

1000

*1000

/MESSAGES

1000	0215	MESS1, 215	/CH
1001	0212	212	/LT
1002	0324	324	/I
1003	0310	310	/H
1004	0305	305	/E
1005	0240	240	/SP
1006	0306	306	/F
1007	0317	317	/U
1010	0314	314	/L
1011	0314	314	/L
1012	0317	317	/U
1013	0327	327	/W
1014	0311	311	/I
1015	0316	316	/N
1016	0307	307	/G
1017	0240	240	/SP
1020	0325	325	/U
1021	0316	316	/N
1022	0305	305	/E
1023	0330	330	/X
1024	0320	320	/P
1025	0305	305	/E
1026	0303	303	/G
1027	0324	324	/I
1030	0305	305	/E
1031	0304	304	/U
1032	0240	240	/SP
1033	0305	305	/E
1034	0322	322	/H
1035	0322	322	/H
1036	0317	317	/U
1037	0322	322	/H
1040	0323	323	/S
1041	0240	240	/SP
1042	0317	317	/U
1043	0303	303	/G
1044	0303	303	/G
1045	0325	325	/U
1046	0322	322	/H
1047	0322	322	/H
1050	0305	305	/E
1051	0304	304	/U
1052	0212	212	/I
1053	0215	215	/CH
1054	0212	212	/LT
1055	0377	377	/RU

1056	0315	MESS2,	315	/M
1057	0301		301	/A
1060	0322		322	/R
1061	0313		313	/K
1062	0240		240	/SP
1063	0324		324	/I
1064	0322		322	/R
1065	0301		301	/A
1066	0303		303	/C
1067	0313		313	/K
1070	0215		215	/CH
1071	0212		212	/LT
1072	0377		377	/RU
1073	0305	MESS3,	305	/E
1074	0316		316	/N
1075	0304		304	/U
1076	0240		240	/SP
1077	0332		332	/E
1100	0317		317	/U
1101	0316		316	/N
1102	0305		305	/E
1103	0215		215	/CH
1104	0212		212	/LT
1105	0377		377	/RU
1106	0323	MESS4,	323	/S
1107	0305		305	/E
1110	0314		314	/L
1111	0305		305	/E
1112	0303		303	/C
1113	0324		324	/I
1114	0215		215	/CH
1115	0212		212	/LT
1116	0377		377	/RU
1117	0320	MESS5,	320	/P
1120	0301		301	/A
1121	0322		322	/R
1122	0311		311	/I
1123	0324		324	/I
1124	0331		331	/Y
1125	0215		215	/CH
1126	0212		212	/LT
1127	0377		377	/RU

1130	0324	MESS6,	324	/I
1131	0311		311	/I
1132	0315		315	/M
1133	0311		311	/I
1134	0316		316	/N
1135	0307		307	/G
1136	0215		215	/CK
1137	0212		212	/LT
1140	0377		377	/RU
1141	0215	MESS7,	215	/CK
1142	0212		212	/LT
1143	0315		315	/M
1144	0305		305	/E
1145	0315		315	/M
1146	0317		317	/U
1147	0322		322	/R
1150	0351		351	/Y
1151	0240		240	/SP
1152	0306		306	/F
1153	0311		311	/I
1154	0305		305	/E
1155	0314		314	/L
1156	0304		304	/U
1157	0240		240	/SP
1160	0305		305	/E
1161	0322		322	/R
1162	0322		322	/R
1163	0317		317	/U
1164	0322		322	/R
1165	0215		215	/CK
1166	0212		212	/LT
1167	0322		322	/R
1170	0311		311	/I
1171	0307		307	/G
1172	0310		310	/M
1173	0324		324	/I
1174	0240		240	/SP
1175	0327		327	/W
1176	0322		322	/R
1177	0317		317	/U
1200	0316		316	/N
1201	0307		307	/G
1202	0215		215	/CK
1203	0212		212	/LT
1204	0377		377	/RU

1205	0215	MESSB,	215	/GR
1206	0212		212	/LF
1207	0304		304	/U
1210	0301		301	/A
1211	0324		324	/I
1212	0301		301	/A
1213	0240		240	/SP
1214	0305		305	/E
1215	0322		322	/R
1216	0322		322	/R
1217	0317		317	/U
1220	0322		322	/R
1221	0215		215	/GR
1222	0212		212	/LF
1223	0306		306	/F
1224	0311		311	/I
1225	0305		305	/E
1226	0314		314	/L
1227	0304		304	/U
1230	0240		240	/SP
1231	0377		377	/RU

1232	0215	MESSY,	215	/GR
1233	0212		212	/LF
1234	0306		306	/F
1235	0311		311	/I
1236	0322		322	/R
1237	0323		323	/S
1240	0324		324	/F
1241	0240		240	/SP
1242	0302		302	/B
1243	0314		314	/L
1244	0317		317	/U
1245	0303		303	/G
1246	0313		313	/K
1247	0240		240	/SP
1250	0377		377	/RU

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1251	0215	MESS10, 215	/CK	
1252	0212	212	/LP	
1253	0314	314	/L	
1254	0317	317	/U	
1255	0303	303	/C	
1256	0256	256	/,	
1257	0240	240	/SP	
1260	0304	304	/U	
1261	0301	301	/A	
1262	0324	324	/I	
1263	0301	301	/A	
1264	0215	215	/CK	
1265	0212	212	/LP	
1266	0377	END, 377		/KO

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RE ARE NO ERRORS

SYMBOL TABLE

BLOCK	0020
ENTR	0021
ERRUR	0022
FIELD	0023
00002	0024
00003	0025
00004	0026
00007	0027
00010	0030
00070	0031
00130	0032
00150	0033
00200	0034
00201	0035
00207	0036
00212	0037
00215	0040
00240	0041
00260	0042
00400	0043
00600	0044
00610	0045
01777	0046
02670	0047
04000	0050
07000	0051
07401	0052
07554	0053
07555	0054
07774	0055
MAX	0056
MESS1	0057
MESS2	0060
MESS3	0061
MESS4	0062
MESS5	0063
MESS6	0064
MESS7	0065
MESS8	0066
MESS9	0067
MES10	0070
INTR1	0071
INTR2	0072
SKIP	0073
STOP	0074
TEMP	0075
INIT	0076
1	0077
2	0100
3	0101
4	0102
5	0103
6	0104

SYMBOL TABLE

Z/	0105
MSPHNT	0106
TYPE	0117
CHLF	0124
PRINT	0135
BELL	0144
NUMBER	0154
BEGIN	0200
START	0224
BERRON	0327
HALT2	0351
HALT5	0353
EMR	0366
SET	0400
STORE	0416
CHECK	0437
DATERM	0462
HALT3	0527
SEARCH	0600
FOUND	0645
WAIT	0651
HALT1	0735
MESS1	1000
MESS2	1050
MESS3	1073
MESS4	1100
MESS5	1117
MESS6	1130
MESS7	1141
MESS8	1205
MESS9	1232
MESS10	1251
END	1266
CUF	6201
DIRA	6761
DTCA	6762
DIRA	6764
DTLA	6766
DTSF	6771
DIRB	6772
DTLB	6774

SYMBOL TABLE

BEGIN	0200
BELL	0144
BERROR	0327
BLOCK	0020
BUF	0201
CHECK	0437
CNTR	0021
CRLF	0124
DATERH	0462
JICA	6762
JILA	6766
JILB	6774
JIRA	6761
JIRB	6772
JISF	6771
JIXA	6764
END	1200
JRR	0366
JRROR	0022
JELLU	0023
UUND	0645
IALIS	0353
IALI1	0735
IALI2	0351
IALI3	0527
00002	0024
00003	0025
00004	0026
00007	0027
00010	0030
00070	0031
00130	0032
00150	0033
00200	0034
00201	0035
00207	0036
00212	0037
00215	0040
00240	0041
00260	0042
00400	0043
00600	0044
00610	0045
1777	0046
2670	0047
4000	0050
7000	0051
7401	0052
7754	0053
7755	0054
7774	0055
MAX	0056
LESS1	1000

SYMBOL TABLE

MESS10	1251
MESS2	1056
MESS3	1073
MESS4	1100
MESS5	1117
MESS6	1130
MESS7	1141
MESS8	1205
MESS9	1232
MSPNN1	0106
NUMBER	0154
PMESS1	0057
PMESS2	0060
PMESS3	0061
PMESS4	0062
PMESS5	0063
PMESS6	0064
PMESS7	0065
PMESS8	0066
PMESS9	0067
PMESS10	0070
PNTK1	0071
PNTK2	0072
PRINT	0135
SEARCH	0600
SET	0400
SKIP	0073
SIANT	0224
SIOF	0074
SIORE	0416
TEMP	0075
TYPE	0117
UNIT	0076
WAIT	0651
Z1	0077
Z2	0100
Z3	0101
Z4	0102
Z5	0103
Z6	0104
Z7	0105